WARNING: This material contains information affecting the national defense of the United States within the meaning of the Espionage Laws, Title 18, U.S.C., Section 793 and 794, the transmission or revelation of which in any manner to an unauthorized person is prohibited by law.

MIL-R-6993A (AER)
15 AUGUST 1955

Supersedes MIL-R-6993(AER)
1 September 1950

MILITARY SPECIFICATION

RADAR SET: AN/APS-20E

This specification has been approved by the Bureau of Aeronautics, Department of the Navy

1. SCOPE AND CLASSIFICATION

1.1 Scope - The equipment covered by this specification shall be primarily an airborne early warning search radar with 360 degree scan for use in military aircraft.

1.2 Classification - The equipment covered by this specification shall be of one type and shall consist of the following:

Major Unit	Type Designation	Max. Allow. Size W (in.)	H (in.)	D (in.)	Max. Allow. Weight lbs.	Applicable Paragraph
Antenna	AS-407/APS-20B	94-3/8	57-3/8	46	180	3.3.1
Duplexer	CU-165/APS-20B	6-1/16	16-1/4	29-1/2	25	3.3.2
Receiver, Radar	R-251/APS-20B	12	12	18	40	3.3.3
Power Supply	PP-829/APS-20B	20	19-5/8	23-1/2	254	3.3.4
Transmitter, Radar	T-467/APS-20E	24-1/2	22-13/16	37-1/4	440	3.3.5
Synchronizer, Electrical	SN-55/APS-20B	15-3/4	11-1/32	22-1/8	55	3.3.6
Indicator, Azimuth and Range	IP-203A/APS-20B	19-13/16	12-1/4	21-3/16	70	3.3.7
Control, Radar Set	C-1449/APS-20E	11-1/8	10-5/8	11-1/4	30	3.3.8
Power Supply	PP-347/APS-20B	15-5/8	15-7/16	22-7/8	65	3.3.9
Range-Azimuth Indicator	IP-23/APS-20B	5-11/16	6-1/4	13/1/8	15	3.3.10
Power Supply	PP-704/APS-20B	10-5/16	7-1/4	14-3/8	25	3.3.11
Transformer Assembly	TF-158/APS-20B	9-3/8	9	33-1/16	80	3.3.12
Coupler, Directional	CU-316/APS-20B	5-1/8	4-1/4	11-3/8	4	3.3.13
Gyroscope	CN-219A/APS-20B	7	12-3/4	9-3/4	14	3.3.14
Gyroscope	CN-218/APS-20B	4-3/16	4	4-9/32	2	3.3.15
Compressor, Air	HD-126/APS-20B	4-9/16	9-3/4	9-3/4	12	3.3.16
Dehydrator	HD-127/APS-20B	6-1/8	3-1/4	10-1/4	4	3.3.17
Cooler, Liquid	HD-125/APS-20B	10-1/8	11-1/4	15-15/16	32.5	3.3.18
Electron Tube Amplifier, Electronic Control	AM-839/APS-20B	4-7/8	7-5/8	6-1/2	4.5	3.3.19
Amplifier, Magnetic	AM-767/APS-20B	4-9/16	6-13/16	6-21/32	8	3.3.20
Mountings						
Mounting	MT-802/APS-20B	25-3/16	2-15/16	19-1/2	10	3.3.2.2 3.3.3.2 3.3.19.2 3.3.20.2
Mounting	MT-954/APS-20B	10-1/4	2-1/4	15-1/4	3	3.3.11.2
Mounting	MT-1134/APS-20B	15-5/8	2-25/32	22-3/4	4	3.3.9.2
Mounting	MT-1135/APS-20B	18-3/8	2-3/4	21-9/16	4.5	3.3.7.2
Mounting	MT-1136/APS-20B	10-15/16	2-13/32	8-1/2	2	3.3.8.2
Mounting	MT-1137/APS-20B	10-7/8	2-5/16	18-5/16	3	3.3.18.2
Mounting	MT-1168/APS-20B	15-5/8	2-25/32	22-3/4	3	3.3.6.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

1.3 Associated Equipment - This equipment shall operate with the following equipments which shall not be supplied as a part of the equipment covered by this specification.

(1)	AN/APA-16	Radar Compensating Assembly
(2)	AN/APX-7	Radar Recognition Set
(3)	AN/APA-57B	Ground Position Indicator
(4)	AN/ART-28	Radio Transmitting Set
(5)	AN/APA-56	Indicator Assembly
(6)	OA-492/APS-20B	Antenna Group
(7)	AN/ARR-26	Radio Receiving Set

2. APPLICABLE DOCUMENTS

2.1 General - The following specifications, standards, publications and drawings of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

2.2 Specifications:

Military

JAN-T-152	Treatment, Moisture and Fungus Resistant, of Communications, Electronic and Associated Electrical Equipment, General Process for
JAN-P-658	Packaging and Packing of Electrical Equipment and Spare Parts (Electronic, Electrical and Electromechanical)
MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (For use with Electronic Equipment in Aircraft)
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Limits and Tests; Aircraft Electrical and Electronic Equipment
MIL-P-7788	Plate, Plastic, Cockpit and Interior Controls Lighting
MIL-E-7894	Electric Power, Aircraft Characteristic of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-N-18307	Nomenclature and Nameplates for Electronic and Associated Equipment
MIL-T-18303	Test Procedures, Preproduction and Inspection Format for

Bureau of Aeronautics

EI-237	Radar Set AN/APS-20E, Installation of

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

2.3 Standards

MIL-STD-129	Marking for Shipment and Storage

2.4 Drawings

Air Force-Navy Aeronautical

AND 10461	Connections Input Power Airborne Electronic Equipment, 28V DC-115V single phase AC and/or 115/220V three phase AC

2.5 Availability of Specifications, Publications and Drawings:

(1) When requesting specifications, publications and drawings refer to both title and number. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable specifications and publications may be obtained upon application to the Commanding Officer, Naval Air Station, Johnsville, Pennsylvania, Attn: Technical Records Division.

(3) Copies of applicable Bureau of Aeronautics drawings and the U. S. Navy Reliable Tube List may be obtained upon application to Bureau of Aeronautics, Navy Department, Washington 25, D. C., Attn: Electronics Division.

2.6 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply.

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the Bureau of Aeronautics.

(3) Reference Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of Parts and Materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following factors shall govern:

(1) Parts and Materials as approved by specifications MIL-E-5400 shall be given first consideration.

(2) When the contractors can demonstrate that the use of standard AN or JAN parts or material will not fulfill the design objectives because of size, weight, performance or other reasons, materials and parts shall be used which most nearly meet or exceed the requirements of the respective standard specification.

(3) When a non-standard part is used, the meeting of general material requirements such as non-inflammability, fungus resistance, non-toxicity, etc. and, the meeting of environmental conditions as required by specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed so that the unit as a whole will pass material, electrical and environmental tests, without consideration to the quality of the detail parts. Any assembly built to be expended shall first be approved by the Bureau of Aeronautics.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

3.1.1 Non-Standard Part Approval and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-E-5400 except as follows:

(1) Mechanical parts are expected to meet the material requirements of MIL-E-5400, and shall be submitted as non-standard parts only when the material requirements are not met.

(2) Non-standard parts, as defined in specification MIL-E-5400 of an electrical nature must be approved before the preproduction model of the equipment is delivered to the Navy. The request for approval must be forwarded by the contractor before the part is purchased or fabricated. In addition to or in lieu of the Non-Standard Part Data Sheet the contractor is requested to submit itemized lists of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

3.1.2 Design Objectives - Minimum size and weight, simplicity of operation and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives in the design of the equipment specified herein. The use of materials, parts and processes other than those required by Specification MIL-E-5400 shall be investigated and when it appears that a substantial reduction in size and weight or improvement in simplicity of design, performance, or reliability can be realized by their use, a request for approval for a non standard part shall be submitted to the Bureau of Aeronautics for consideration. Each request should be accompanied by complete supporting information.

3.1.3 Plug-in Assemblies - The use of easily disconnected plug-in assemblies of one or more related circuits shall be investigated with a view towards reduction of size and weight, simplification of operation and maintenance or improvement in performance.

3.1.4 Choice of Tube Types - Electronic tubes used in the equipment shall be chosen in accordance with the requirements of MIL-E-5400. However where there is a counterpart listed in the U. S. Navy Reliable Tube List (copies may be obtained from the Bureau of Aeronautics) the reliable version shall be used. The use of any tube not on these acceptable listings must be approved by the Bureau of Aeronautics; however, before consideration for approval, the contractor must submit the following information:

(1) Quantitative data on the electrical advantages and mechanical adequacy of the proposed tube over the most nearly equivalent type on the acceptable listings.

(2) Test data and performance limits of the tube which must be maintained to insure satisfactory and reliable performance in the equipment.

(3) The tube manufacturer shall have received Military or Navy qualification approval for that particular tube.

3.1.5 Workmanship - Workmanship, fabrication and manufacturing processes shall conform to the requirements of MIL-E-5400.

3.2 General Design Requirements

3.2.1 General - This equipment shall fulfill all applicable design and performance requirements of Specification MIL-E-5400 unless otherwise specified herein.

3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions or reasonable combinations of service conditions as specified in Specification MIL-E-5400 with the following exceptions:

(1) Sand and Dust - The equipment shall be designed to be as dust proof as practicable consistent with cooling requirements. Testing of the equipment in a turbulent atmosphere filled with sand and dust concentrations will not be required of the contractor.

(2) Vibration - The vibration requirement shall apply except the double amplitude value of 0.06 inch shall be changed to 0.03 inch when applied to the following units:

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

AS-407/APS-20B	Antenna
TF-158/APS-20B	Transformer Assembly

Vibration with isolators removed in accordance with MIL-E-5400 shall not be required.

3.2.3 Total Weight - The total weight of the equipment excluding cables and hose assemblies shall not exceed 1500 pounds.

3.2.4 Operational Stability - After 20 minutes warm up time, the equipment shall operate with optimum performance for 24 hours, continuously or intermittently, without necessity for readjustment of any controls which are non-accessible during flight.

3.2.5 Operating Life

3.2.5.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 500 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.5.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

3.2.6 Interconnecting Cabling - This equipment shall be capable of satisfactory operation using external wiring in accordance with Specification MIL-W-5088. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

3.2.7 Cables and Connectors - The equipment shall be designed for the use of cables and connectors in accordance with the requirements of Specification MIL-E-5400. However, cables and parts of connectors which are to be attached to the cables shall not be supplied as part of the equipment.

3.2.8 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power sources with characteristics as defined in MIL-E-7894 having limits as specified herein. The power required shall not exceed the specified amounts.

(1) AC Power (Single Phase)

(a) 115 Volt Constant Frequency

Operating Voltage Limits	110 to 120 Volts
Operating Frequency Limits	380 to 420 cps
Reduced Performance Voltage Limits	108 to 121 Volts
Reduced Performance Frequency Limits	360-380 and 420-440 cps
Power (1φ)	250 V. A.

(2) AC Power(Three Phase)

(a) 115/200 Volt Variable Frequency

Operating Voltage Limits	110 to 120 Volts
Operating Frequency Limits	380 to 1000 cps
Reduced Performance Voltage Limits	108 to 121 Volts
Reduced Performance Frequency Limits	380 to 1000 cps
Power (3φ)	7500 V. A.

(3) DC Power 28 Volts System

Operating Voltage Limits	24 to 29 Volts
Current	44 Amps

3.2.9 Input Power Connections - Input power connections shall be in accordance with drawing AND-10461.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

3.2.10 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3.2.10.1 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within limits.

3.2.11 Standby Operating Provisions - The equipment shall be designed to provide for a standby position whereby rated voltages shall be applied to all tube filaments.

3.2.12 Moisture and Fungus Resistant Treatment - Equipments shall be fungus proofed by selection of parts and materials as non-nutrient for fungus or the parts and materials shall be treated prior to their use in the equipment so that overall spraying of the equipment is not necessary. Overall spraying of the equipment in accordance with JAN-T-152 shall be made in the event selection of parts and materials described above is not possible.

3.2.13 Identification and Marking - The equipment shall be identified and marked as follows:

3.2.13.1 Nomenclature and Nameplates - Nomenclature assignment and Nameplate approval for equipment identification shall be in accordance with the requirements of specification MIL-N-18307.

3.2.13.2 Special Markings - Any plug-in units and/or other sub-assemblies that do not carry a nameplate shall carry the following special markings as applicable.

(1) The name or acceptable abbreviations of the name of the plug-in or sub-assembly (e.g. I-F Strip, Pre-amplifier, etc.) shall be stamped on the item if space permits.

(2) The letter and number portion of the major component nomenclature, preceded by p/o, shall be stamped on the item (e.g. p/o R-00/APS-00).

(3) The stamping or markings shall meet the requirements of Specification MIL-N-18307.

3.2.14 Interchangeability - The interchangeability requirements as specified in specification MIL-E-5400 shall be extended to require interchangeability of units and component parts of this equipment with the latest model of any previously produced equipments. Such interchangeability shall be measured against a model, manufacturing drawings, or other technical information as provided by the Bureau of Aeronautics for that purpose. In the event that requirements of MIL-E-5400 conflict with the requirements for interchangeability, the interchangeability requirements shall govern.

3.2.15 Provisions for Maintenance and Field Testing - Provisions for maintenance and field testing shall meet the requirements of Specification MIL-E-5400.

3.2.16 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.17 Pressurization - Provisions shall be made for suitable pressurization of Power Supply PP-829/APS-20B and Radar Transmitter T-467/APS-20E to a minimum of 20 psi absolute, and the Duplexer CU-165/APS-20B, Antenna AS-407/APS-20B and all R. F. lines to a minimum of 30 psi absolute. Leakage of the system shall not exceed that required to cause the compressor to operate at a duty cycle in excess of 75% at an altitude of 40,000 feet.

3.2.18 Performance

3.2.18.1 General - The equipment shall be designed as part of the Airborne Early Warning System. It shall be capable of transmitting, receiving and displaying radar signals and of receiving and displaying beacon signals. It shall also be capable of supplying time coded azimuth pulses and video information to a radar relay transmitter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

(1) The following provisions shall be required to operate this equipment with the AN/APA-16.

(a) The AN/APS-20E shall supply a positive trigger, coincident with the transmitter pulse and having a nominal amplitude of 40 volts, to the AN/APA-16.

(b) Azimuth and Range Indicator IP-203A/APS-20B shall accept and display a video tracking marker with a positive polarity and a nominal amplitude of 6 volts supplied by the AN/APA-16. The AN/APS-20E heading marker shall be displayed on Azimuth and Range IP-203A/APS-20B when the AN/APA-16 tracking marker is displayed.

(2) The following provision shall be required to operate this equipment with the AN/APA-57B: A one speed, three phase differential generator shall be provided in Antenna AS-407/APS-20B to accept north stabilized stator voltages from an AY-201-3-B type synchro generator in the AN/APA-57B, insert azimuth rotational information and deliver north stabilized data to Electrical Synchronizer SN-55/APS-20B.

(3) Azimuth and Range Indicator IP-203A/APS-20B shall accept and display a nominal 5 microsecond, 2 volt pulse from the AN/ARR-26 Radio Receiving Set.

(4) The following provisions shall be required for operation with the AN/APA-56 Indicator Assembly.

(a) The cosine pulse, modulator trigger, and Radar and Beacon video shall be provided for the AN/APA-56.

(b) Electrical Synchronizer SN-55/APS-20B shall accept mixed radar and grid map video from the AN/APA-56 and supply this information to the AN/ART-28 Radio Transmitting Set.

(5) The following provisions shall be required for operation with the AN/APX-7 Radar Recognition Set:

(a) Azimuth and Range Indicator IP-203A/APS-20B shall accept and display all three modes of IFF video received from the AN/APX-7.

(b) The IFF Antenna Array mounted on Antenna AS-407/APS-20B shall radiate and receive energy for the AN/APX-7 Radar Recognition Set.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

3.2.18.2 System Noise Figure and Minimum Perceptible Signals - The overall system noise figure shall be less than 9 db. This measurement shall take precedence over sensitivity when determining performance of system. The radar minimum perceptible signal shall be greater than 137 db below one watt and the beacon minimum perceptible signal shall be greater than 120 db below 1 watt.

3.2.18.3 Anti-Jam Provisions - The following anti-jam provisions shall be incorporated in the receiving system:

(1) Fast Time Constant

(2) Sensitivity Time Control & IAGC Controlled I-F Strip

3.2.18.4 Range Sweep Characteristics - Range sweep characteristics shall be as follows:

Indicator	Radar Repetition Rate	Sweep Speed Nauti. Mi. Per Radius	Sweep Duration Nauti. Mi.	Range Marks Nauti. Mi.	Off-setting by Local Slew	G. P. I. Stabili-zation	Azimuth Stabili-zation
IP-203A/APS-20B	300	10	50	10	Yes	Yes	True
IP-203A/APS-20B	300	20	80	10	Yes	Yes	True
IP-203A/APS-20B	300	50	200	10	Yes	Yes	True
IP-203A/APS-20B	300	100	200	20	Yes	Yes	True
IP-203A/APS-20B	300	200	200	20	Yes	Yes	True
IP-203A/APS-20B	900	10	50	10	Yes	Yes	True
IP-203A/APS-20B	900	20	50	10	Yes	Yes	True
IP-203A/APS-20B	900	50	50	10	Yes	Yes	True
IP-203A/APS-20B	900	100	50	20	Yes	Yes	True
IP-203A/APS-20B	900	200	50	20	Yes	Yes	True
IP-23/APS-20B	300	50	50	20	No	No	Relative
IP-23/APS-20B	300	150	150	20	No	No	Relative
IP-23/APS-20B	900	50	50	10	No	No	Relative

3.2.18.5 Azimuth and Range Accuracy - Indicated azimuth on IP-203A/APS-20B shall be within 2 degrees of true or relative bearing of the antenna as applicable exclusive of errors in associated compass systems. Indicated range accuracy shall be within plus or minus 2 percent at maximum range.

3.2.18.6 Minimum Range Performance - Minimum range shall depend on recovery time of the duplexer which shall be within a curve defined by the following points plotted on linear graph paper; 3 db at 36 usec, 10 db at 20 usec, 25 db at 10 usec.

3.2.18.7 Maximum Range Performance - This equipment should be capable of detecting targets at the following approximate ranges in the absence of clutter.

Target	Range
2000 Ton Destroyer	200 nautical miles
Fighter Aircraft	65 nautical miles
Medium Bomber Aircraft	85 nautical miles

3.2.18.8 Frequency Characteristics - This equipment shall be designed for pulsed radar operation at 2880 ± 30 megacycles per second and shall also be capable of receiving radar beacon transmission at a frequency of 2820 ± 1 megacycles per second.

3.2.18.9 Modulation Characteristics - The transmitter shall be pulse modulated at a pulse repetition rate of 300 ± 30 cycles per second or 900 ± 90 cycles per second. At the 300 cycle rate, the modulated pulse will have a nominal width of 2 microseconds. At the 900 cycle rate the modulated pulse will have a nominal width of 0.67 microsecond.

3.2.18.10 Transmitter Power Output - The equipment shall be capable of delivering not less than 1.75 megawatts

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

3.3 Detail Design Requirements

3.3.1 Antenna AS-407/APS-20B

3.3.1.1 Function - The antenna shall provide for focusing, radiating, and receiving radar and IFF signals and for receiving beacon signals, and shall provide azimuth position information to Electrical Synchronizer SN-55/APS-20B and Range-Azimuth Indicator IP-23/APS-20B.

3.3.1.2 Form Factor and Mounting - Antenna AS-407/APS-20B shall not exceed the dimensions shown in Bureau of Aeronautics Drawing EI-237-1 and 2. The assembly shall be supported by the gear case mounting feet.

3.3.1.3 Weight - The weight of Antenna AS-407/APS-20B shall not exceed 180 pounds.

3.3.1.4 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

3.3.1.5 Pressurization - All r-f lines and components shall be pressurized to a minimum of 30 psi absolute and shall be capable of withstanding a maximum pressure of 40 psi absolute. Leakage shall not exceed that specified in paragraph 3.2.17.

3.3.1.6 Power Capacity - When looking into free space, the peak power capacity of all radar r-f components shall be not less than 3 megawatts and all IFF r-f components shall be not less than 6 kilowatts under the conditions of pressurization herein specified.

3.3.1.7 Standing Wave Ratio

3.3.1.7.1 Radar - When looking into free space the voltage standing wave ratio shall be less than 1.25:1 with the antenna in any azimuth position and at all frequencies between 2850 and 2910 megacycles per second.

3.3.1.7.2 IFF - When looking into free space, the voltage standing wave ratio shall be less than 1.7:1 throughout the operating range of 1030 to 1090 megacycles per second and over the tilt range specified for the radar beam.

3.3.1.8 Antenna Gain

3.3.1.8.1 Radar - Antenna gain shall be not less than 30 db relative to an isotropic radiator.

3.3.1.8.2 IFF - Antenna gain shall be not less than 18 db in the direction of the maximum power radiation of the major lobe.

3.3.1.9 Radiation Patterns - Radiation patterns shall have the following characteristics under the specified operating conditions:

3.3.1.9.1 Radar

3.3.1.9.1.1 Polarization - The radiated field shall be horizontally polarized.

3.3.1.9.1.2 Horizontal Pattern - The beam pattern in the horizontal plane shall have half power points at angles not greater than 1.75 degrees either side of the direction of maximum power radiation.

3.3.1.9.1.3 Vertical Pattern - The beam pattern in the vertical plane shall have half power points at angles not greater than 4 degrees either side of the direction of maximum power radiation.

3.3.1.9.1.4 Side Lobes - All side lobes in the E plane shall be at least 17 db below the main lobe. All side lobes in the H plane shall be at least 20 db below the main lobe.

3.3.1.9.1.5 Tilt - At any azimuth position, the axis of the vertical beam shall be capable of tilting from +15 degrees to -15 degrees with respect to a plane through the mounting surfaces of the assembly. Distortion of the horizontal pattern resulting from such tilt shall be held to less than two db loss in gain, and less than one degree increase in either the vertical or horizontal radar beamwidth.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

3.3.1.9.2 IFF

3.3.1.9.2.1 Polarization - The radiated field shall be vertically polarized.

3.3.1.9.2.2 IFF and Radar Beam Alignment - Under all conditions of specified operation, the line of maximum power radiation in azimuth of the IFF beam shall not deviate by more than 0.5 degree from the line of maximum power radiation in azimuth of the radar beam. Under all conditions of specified operation, the IFF radiation in the vertical plane shall not be less than 6 db down from its line of maximum power radiation for all angles for which the radar radiation in the vertical plane is not less than 3 db down from its line of maximum power radiation.

3.3.1.9.2.3 Horizontal Pattern - The beam pattern in the horizontal plane shall be not greater than 12 degrees in overall width between the one-half power points.

3.3.1.9.2.4 Continuity of Pattern - The major lobe of the radiated pattern shall contain no sharp minima of radiated power. Transistion from maximum radiation to lower levels of radiated energy shall be a smooth curve free of discontinuities to at least 20 db below maximum.

3.3.1.9.2.5 Side Lobes - All side lobes in the E plane shall be at least 18 db below the main lobe. All side lobes in the H plane shall be at least 13 db below the main lobe.

3.3.1.10 Antenna Scan

3.3.1.10.1 Rotation - The scanning mechanism shall be capable of rotating the radiated beams continuously through 360 degree clockwise rotation at a slow speed of 2.4 rpm plus or minus 10%, or a fast speed of 6 rpm plus or minus 10%. By means of a gear change in the driving assembly, alternate speeds of 6 and 15 rpm plus or minus 10% shall be provided.

3.3.1.10.2 Control - Suitable relay circuits shall be provided in the Antenna Assembly to permit stopping, starting or sector scanning of the antenna from Radar Set Control C-1449/APS-20E except that sector scanning at 15 rpm shall not be required. On sector scan the antenna shall be capable of operating continuously with a sector width of ±20 to ±55 degrees about the selector sector center.

3.3.1.11 Antenna Tilt - Vertical tilt of the radiated beams shall be accomplished by positioning the reflector with respect to the radar feed.

3.3.1.12 Rotating Joint - A dual frequency rotating joint shall be provided to accomplish the transfer of radar, beacon and IFF energy between the rotating antenna assembly and the stationary transmission lines running to the radar and AN/APX-7 IFF equipment.

3.3.1.12.1 Attenuation - Energy within the designated frequency range of each section of the rotating joint shall not be attenuated by more than 0.1 db in transfer.

3.3.1.13 Interference between Radar and IFF - There shall be no coupling of energy between the r-f components of the antenna which will produce a detectable signal in the radar receiving equipment. There shall be at least 40 db rejection of signals in the frequency range from 2850 to 2910 megacycles at the input of the IFF channel.

3.3.14 Antenna Tilt Stabilization

3.3.1.14.1 Tilt Stabilization Operation - Under stabilized operation, the elevation angle of the radar beam shall be stabilized with respect to the horizontal within an accuracy of ±2 degrees. This accuracy shall be maintained for tilt angles of 15 degrees or less, tilt rates of 32 deg/sec or less and tilt accelerations of 50 deg/sec/sec or less.

3.3.1.14.2 Tilt Unstabilized Operation - Under unstabilized operation, the reflector tilt angle shall be controlled relative to the aircraft horizontal plane. The elevation angle of the radar beam shall correspond to the setting of the manual tilt control through an angle of ±15 degrees within an accuracy of ±1 degree.

3.3.14.3 Tilt Limit - For tilt angles greater than ±15 degrees tilt stabilization shall be inoperative, but the equipment shall not be damaged and shall return to satisfactory operation when the aircraft returns within stabilization limits.

3.3.1.14.4 Stow - Means shall be provided to lock the antenna reflector when the Tilt Selector Switch is turned to STOW.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

3.3.1.14.5 Tilt Information - Means shall be provided in the radar set control to transmit the angle of tilt of the radiated beam with an accuracy of ±0.5 degree to one or more remote locations.

3.3.1.15 Azimuth Information to Range-Azimuth Indicator IP-23/APS-20B - A one speed three phase sweep resolver whose rotor is mechanically linked to Antenna AS-407/APS-20B rotation shall accept sweep voltages generated in Power Supply PP-704/APS-20B and provide relative azimuth information in the form of modulated output voltage to Range-Azimuth Indicator IP-23/APS-20B.

3.3.2 Duplexer CU-165/APS-20B

3.3.2.1 Function - The duplexer shall be a polarization shifting type utilizing quartz gas filled TR tubes. It shall provide for transmission of radar r-f energy and reception of radar and beacon signals.

3.3.2.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown in Bureau of Aeronautics Drawing E1-237-7. It shall mount on Mounting MT-802/APS-20B.

3.3.2.3 Weight - The weight of the duplexer shall not exceed 25 pounds.

3.3.2.4 Electrical Connections - External electrical connections shall be made in accordance with Bureau of Aeronautics Drawing E1-237-36.

3.3.2.5 Power Capacity - The duplexer shall be capable of carrying at least three megawatts of transmitted peak power when pressurized.

3.3.2.6 Pressurization - The duplexer shall be pressurized to a minimum of 30 psi absolute and shall be capable of withstanding pressure of 40 psi absolute. Leakage shall not exceed that specified in paragraph 3.2.17.

3.3.2.7 Standing Wave Ratio - The high level VSWR shall be less than 1.15:1 throughout the frequency range of 2850 to 2910 megacycles per second.

3.3.2.8 TR Tubes - The duplexer TR quartz tube shall be 3 mm OD, filled with alumina chips or equivalent at 50 mm gas pressure. Additional TR protection shall be furnished by a type 6117 tube in the receive arm of the duplexer.

3.3.2.9 Transmit Losses - The loss in the duplexer on transmit shall not exceed 0.4 db.

3.3.2.10 Radar Signal Mixer - It shall be a balanced crystal mixer employing two type 1N21C crystals.

3.3.2.11 Radar AFC Pick-up Loop - The AFC attenuator shall provide proper attenuation for the radar AFC mixer.

3.3.2.12 Beacon Signal Mixer - It shall utilize a filter cavity tuned to the beacon frequency and a type 1N21B crystal.

3.3.2.13 Crystal Shutter - R-f energy transmitted through the TR tubes shall be attenuated at least 35 db when the shutter is closed. This shutter shall open automatically when the system power switch is in the OPERATE position and the system has timed out.

3.3.3 Radar Receiver R-251/APS-20B

3.3.3.1 Function - The receiver shall be capable of receiving intermediate frequency signals from the radar and beacon signal mixers, amplifying and detecting them, and furnishing video signals at the radar and beacon outputs. It shall supply the keep-alive voltage for the TR tube. It shall contain the local oscillators and AFC circuits for the radar and beacon channels.

3.3.3.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing E1-237-8. It shall mount on Mounting MT-802/APS-20B.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

3.3.3.3 Weight - The weight of Radar Receiver R-251/APS-20B shall not exceed 40 pounds.

3.3.3.4 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

3.3.3.5 Controls - The location, lettering and type of all controls shall be in accordance with Bureau of Aeronautics Drawings EI-237-8 and EI-237-21.

3.3.3.6 Radar Receiver Performance

3.3.3.6.1 Intermediate Frequency Amplifier - The radar i-f amplifier shall consist of a low noise pre-amplifier, and an i-f amplifier strip with STC and IAGC controlled stages. Design of the intermediate frequency amplifier shall be such as to provide two i-f outputs (normal and AMTI). The AMTI shall be obtained by taking the i-f ahead of the final IAGC controlled stage and following with a linear i-f stage and detector.

3.3.3.6.1.1 I-F Amplifier Frequency - The amplifier shall be tuned to 30 ± 0.25 mc/sec.

3.3.3.6.1.2 Response Characteristics - By means of IAGC feedback maintenance adjustment, it shall be possible to compress a signal 80 db above noise to 1/2 of the amplitude of the initial spike. At this IAGC adjustment, it shall be possible to set the noise amplitude at 1/4 of the amplitude of the spike.

3.3.3.6.1.3 I-F Bandwidth - I-f bandwidth shall be 1.2 mc per sec plus or minus 0.2 mc per sec measured between half-power points on normal radar channel and 1.4 mc per sec plus or minus 0.3 mc per sec on AMTI channel.

3.3.3.6.1.4 Sensitivity - The signal to noise ratio shall be such as to produce a minimum perceptible signal level of 137 db below one watt.

3.3.3.6.1.5 STC - With the IAGC potentiometer set for two to one compression, the radar gain control adjusted to maximum noise and the suppressor bias level set to bring the noise level to 1/4 the amplitude of a saturated signal, the STC adjust control shall be capable of reducing the receiver sensitivity a minimum of 70 db at five miles. With the STC adjust set for 70 db sensitivity reduction at five miles, the sensitivity shall recover to 3 db below maximum at 80 ± 20 radar miles.

3.3.3.6.2 Radar LO and AFC Circuits

3.3.3.6.2.1 Characteristics - The local oscillator shall be maintained at a frequency 30 mc per sec above the transmitter frequency. The LO and AFC circuits shall be designed for reception of signals from 2850 to 2910 mc per sec.

3.3.3.6.2.2 Radar AFC Mixer - The radar AFC mixer shall accept the output of the radar AFC pickup loop and shall employ a 1N21B crystal. A coaxial attenuator shall be provided to attenuate magnetron harmonic frequencies. The attenuation shall be 15 ± 1 db at 2880 mc.

3.3.3.6.3 Video Outputs

3.3.3.6.3.1 Video Output Characteristics - Outputs shall be provided for RADAR, BEACON and AMTI videos. These videos shall be provided by a linear source and shall be of a nominal two volt peak amplitude when shunted by 500 uuf capacitance. The output level shall be adjustable to an amplitude of 2 volts ± 10%.

3.3.3.6.3.2 Video Bandwidth - The video channels for RADAR, BEACON, and AMTI shall pass video pulses from 0.5 to 1000 microseconds duration with a maximum rise time of 0.25 microsecond and a maximum "droop" of 1.0 db.

3.3.3.7 Beacon Receiver Performance

3.3.3.7.1 Intermediate Frequency Amplifier

3.3.3.7.1.1 I-F Frequency - The beacon i-f amplifier shall be tuned to 40 ±1.0 mc per sec.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

3.3.3.7.1.2 Bandwidth - The bandwidth at the half-power points shall not be less than 6 mc per sec. nor more than 8 mc per sec at the -3 db points, and, at the -10 db points, it shall be greater than 16 mc per sec.

3.3.3.7.1.3 Input - The input shall be designed to operate from a single-ended mixer using a type 1N21B crystal.

3.3.3.7.1.4 Sensitivity - The sensitivity shall be such as to produce a minimum perceptible signal of at least 120 db below one watt.

3.3.3.7.2 Beacon LO and AFC Circuits

3.3.3.7.2.1 Characteristics - Provision shall be made to automatically hold the frequency of the local oscillator to within ±0.5 mc per sec of the frequency of a reference cavity. The reference cavity shall maintain its set frequency within ±0.5 mc per sec and shall be delivered adjusted for beacon reception at 2820 mc per sec but shall be capable of adjustment for beacons between 2730 and 2830 mc per sec.

3.3.3.7.3 Video Amplifier

3.3.3.7.3.1 Video Bandwidth - It shall be capable of amplifying pulses from 0.5 to 100 microseconds duration with a maximum rise time and "droop" of 0.25 microsecond and 1.0 db respectively when shunted by 500 uuf capacitance.

3.3.3.7.3.2 Video Output Characteristics - It shall be designed to provide a 2 volt peak output signal when shunted by 500 uuf capacitance. Provisions shall be included to limit the video above a fixed level.

3.3.4 Power Supply PP-829/APS-20B

3.3.4.1 Function - This unit shall be designed to accept the three-phase voltage output from a variable auto-transformer in Transformer Assembly TF-158/APS-20B, rectify it and supply the required d-c voltage to Radar Transmitter T-467/APS-20E.

3.3.4.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown in Bureau of Aeronautics Drawing EI-237-17. It shall be mounted by means of self-contained vibration mounts.

3.3.4.3 Weight - The weight of this unit shall not exceed 250 pounds.

3.3.4.4 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

3.3.4.5 Pressurization - The unit shall be pressurized at a minimum of 20 psi absolute and shall be capable of withstanding a pressure of 40 psi absolute. Pressure tightness shall be achieved by means of a "hat" and a clamp-ring arrangement. Leakage shall not exceed that specified in paragraph 3.2.17.

3.3.4.6 Output Voltage, Current and Ripple - At 110 volt line to neutral input the output must exceed 12 kv at 300 ma. The rms ripple shall not exceed 1% of the d-c output voltage.

3.3.5 Radar Transmitter T-467/APS-20E

3.3.5.1 Function - The unit shall be designed to accept the high voltage d-c from Power Supply PP-829/APS-20B, convert it to a high voltage driving pulse for the QK428 transmitting tube which shall deliver a high power r-f pulse to the antenna. A positive trigger, coincident in time with the transmitted pulse, shall be provided for test purposes.

3.3.5.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing EI-237-16. It shall be mounted by means of self-contained vibration mounts.

3.3.5.3 Weight - The weight of Radar Transmitter T-467/APS-20E shall not exceed 440 pounds.

3.3.5.4 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

3.3.5.5 Pressurization - The unit shall be pressurized at a minimum of 20 psi absolute and shall be capable of withstanding a pressure of 40 psi absolute. Pressure tightness shall be achieved by means of a "hat" and clamp-ring arrangement. Leakage shall not exceed that specified in paragraph 3.2.17.

3.3.5.6 Protection - The unit shall be provided with a safety interlock that will remove all high voltages when the "hat" is removed.

3.3.5.7 Modulator Section of Transmitter Unit

3.3.5.7.1 Repetition Rate - The modulator section shall be capable of being triggered at a repetition rate of 300 or 900 pulses per second.

3.3.5.7.2 Pulse Shape - The modulator shall deliver a 2.0 microsecond pulse at 300 pps and a 0.67 microsecond pulse at 900 pps. The pulse shall be negative with respect to ground and shall have a rise and fall time consistent with the requirements of a type QK-428 magnetron.

3.3.5.7.3 Pulse Amplitude - The peak amplitude shall be controllable from 60 to 100 percent of the transmitting tube input rating to achieve 1.75 megawatt output, such control to be derived from the variable output of Power Supply PP-829/APS-20B.

3.3.5.8 Transmitter Section - The power tube shall be a type QK-428 magnetron.

3.3.5.8.1 Power Output - The peak power of the transmitted pulse shall be determined by measuring the average power absorption of a matched water load and dividing by the repetition rate times the measured current pulse width at the 50% point. Minimum acceptable peak power output shall be 1.75 megawatts, measured directly at the output coupling of the transmitter.

3.3.5.8.2 Pulse Shape - The envelope of the r-f pulse shall have a width of 1.9 to 2.1 microseconds measured at 50% amplitude for the 300 repetition rate and 0.6 to 0.75 microsecond measured at 50% amplitude for the 900 repetition rate.

3.3.5.8.2.1 Pulse Stability- Delay between the modulator trigger and the transmitter pulse shall not vary more than .01 microsecond from pulse to pulse.

3.3.5.8.3 Transmitter Test Pulse - An auxiliary winding shall furnish a positive trigger coincident with the transmitted pulse. The nominal amplitude of this trigger shall be 40 volts.

3.3.5.8.4 Frequency - Frequency of the transmitted energy shall be 2880 plus or minus 30 megacycles.

3.3.6 Electrical Synchronizer SN-55/APS-20B

3.3.6.1 Function - The synchronizer shall generate a basic pulse for timing the radar functions; furnish a modulating signal containing azimuth and video information to the radar relay transmitter; and contain a servo amplifier, resolvers and synchros to control and distribute azimuth information.

3.3.6.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing EI-237-19. It shall be mounted on Mounting MT-1168/APS-20B.

3.3.6.3 Weight - The weight of Electrical Synchronizer SN-55/APS-20B shall not exceed 55 pounds.

3.3.6.4 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

3.3.6.5 Basic Pulse - The basic pulse shall be of positive polarity and have a nominal peak amplitude of 20 volts across a 100 uuf capacitor. The repetition rate will be selected at the radar set control. The low repetition rate shall be 300 ±30 pps and shall be adjustable at 270 pps by maintenance controls for operation with the AN/APA-56 system. The high repetition rate shall be 900 ± 90 pps. The basic pulse at this repetition rate will trigger the modulator directly. An external jack shall be provided to accept a substitute 900 pps trigger from auxiliary equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

3.3.6.6 Sine Pulse - The sine pulse shall follow the basic pulse by a time equal to a range corresponding to the sum of 30,400 yards plus the product of a constant times the sine of the angle between the axis of the radar beam and true north. The time delay error of the sine pulse shall be less than 4% of the constant (in microseconds).

3.3.6.7 Cosine Pulse - The cosine pulse shall follow the sine pulse by a time equal to a range corresponding to the sum of 30,400 yards plus the product of a constant times the cosine of the angle between the axis of the radar beam and true north. The time delay error of the cosine pulse shall be less than 4% of the constant (in microseconds).

3.3.6.8 Modulator Trigger - The modulator trigger shall be of positive polarity with an amplitude of 40 volts across a 1000 uuf capacitor. It shall be used to trigger the modulator section of Radar Transmitter T-467/APS-20E and sweeps on Indicators IP-23/APS-20B and IP-203A/APS-20B. The modulator trigger shall follow the cosine pulse by 30 ± 3 microseconds on the low repetition rate and on the high repetition rate it shall be coincident with the basic pulse. An emergency-normal switch shall be provided on the synchronizer as a maintenance control to permit the basic pulse to trigger the modulator at the low repetition rate.

3.3.6.9 IFF Trigger - The IFF trigger shall be of positive polarity and have a minimum peak amplitude of 5 volts as measured across a 500 uuf capacitance. The repetition rate shall be the same as that for the transmitter and shall be coincident with the cosine pulse at 300 pps and coincident with the modulator trigger at 900 pps.

3.3.6.10 Relay Azimuth Information - The modulating signal for the radar relay transmitter shall consist of basic, sine, and cosine pulses plus the modulator pulse or the coded IR pulse on the low repetition rate only. There shall be no modulating signal on the high repetition rate. Each pulse shall be transformed into three one microsecond pulses whose leading edges are separated by 3 ±0.3 and 5 ± 0.5 microseconds respectively. This signal shall be a positive polarity with a peak voltage of 5 ± 1 volts as measured across a 500 uuf capacitance.

3.3.6.11 IFF Mode Identification - The IFF mode identification data shall be composed of one of the three IFF mode identification pulses (coded IR pulses) supplied by the AN/APX-7 IFF equipment. This signal will occur at 15, 20, or 25 microseconds after the cosine pulse depending upon the IFF mode of operation selected. The pulses shall be of positive polarity and have a minimum peak amplitude of 10 volts. Each IR pulse shall be transformed into three one microsecond pulses whose leading edges are separated by 3 ± 0.3 and 5 ± 0.5 microseconds respectively. During radar IFF time sharing, the coded IR pulse will replace the modulator pulse on those intervals where only IFF video is relayed.

3.3.6.12 Relay Video - The video data shall be of positive polarity, and adjustable in amplitude from 0 to 2 volts measured across 500 uf capacitance. It shall follow the synchronizing data and consist of radar and/or beacon and/or IFF video. During IFF interrogation the radar and/or beacon video shall alternate on successive cycles with the IFF video. In addition provisions shall be incorporated to permit IFF video to be mixed directly with radar and/or beacon video by selection at the radar set control. Suitable mixing circuits shall be provided to supply the relay transmitter with radar, beacon, and IFF video separately or mixed in any combination. Video selection shall be at the radar set control.

3.3.6.13 Range Marks - Ten and twenty mile range mark circuits shall be provided to furnish range markers to all Range and Azimuth Indicators and to the radar set control. Both ten and twenty mile range markers are to be available simultaneously. The first range marker shall occur not later than 4 microseconds after the modulator trigger. The markers shall continue for a period of approximately 2400 microseconds after the first marker except for the high repetition rate where the markers shall continue for a period of approximately 600 microseconds. The width of the markers shall not exceed one microsecond measured at 85% peak amplitude. The back swing shall not exceed one volt. The markers shall maintain an accuracy in time of ±2%.

3.3.6.14 Heading Mark - A heading marker composed of markers of approximately one mile intervals extending over the same duration as the range markers shall be generated in the Electrical Synchronizer SN-55/APS-20B. They shall be transmitted to Azimuth and Range Indicator IP-203/APS-20B and associated equipments when the antenna passes through the dead ahead position. When sector scanning at 6 rpm about the dead ahead position, the heading markers as observed on an Azimuth and Range Indicator IP-203A/APS-20B shall not be separated by more than one degree.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

3.3.6.15 Servo Drive Assembly - The servo-drive assembly shall be designed to accept true bearing azimuth voltages from a one-speed three-phase synchro differential generator on the antenna amplifying them and apply them to a 400 cycle control motor to position the sweep resolvers for the Azimuth and Range Indicators IP-203A/APS-20B and to position a synchro generator (Type AY-402 or equivalent) for furnishing true bearing azimuth information to the sector scan control circuits, Scott Transformer and associated equipment of the AN/APA-56 Indicator Assembly.

3.3.6.16 Sector Scan - The sector scan circuit shall be designed to cause Antenna AS-407/APS-20B to sector scan about any selected bearing within the limits of ±20 to ±55 degrees. Sector circuitry should prevent Antenna OA-492/APS-20B from sectoring less than ±7.5 degrees. Control of sector center-bearing and width shall be at the radar set control.

3.3.7 Azimuth and Range Indicator IP-203A/APS-20B

3.3.7.1 Function - This unit shall be capable of displaying Radar, Beacon, IFF, AMTI, Sonobuoy, Heading Marker and LAB Tracking Marker video information, on a PPI type presentation, utilizing a 7 inch KRT with P7 persistency, on either a centered or off-centered display. It shall provide for offsetting the necessary range sweep through acceptance of AN/APA-57B Ground Position Indicator output voltages or by local slew voltages. It shall provide a strobe and cursor, accurate azimuth and range determination, provide all operating controls for varying the presentation and contain the high voltage supply for the KRT.

3.3.7.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing EI-237-23. It shall be mounted on Mounting MT-1135/APS-20B.

3.3.7.3 Weight - The weight of the Azimuth and Range Indicator IP-203A/APS-20B shall not exceed 70 pounds.

3.3.7.4 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

3.3.7.5 Controls - The location, lettering, and type of front panel controls shall be in accordance with Bureau of Aeronautics Drawing EI-237-23. The following maintenance controls shall be provided:

Qty.		Qty	Description
2	Cursor Centering Controls (N/S and E/W)	1	Sweep Speed Balance Control
2	PPI Centering Controls (N/S and E/W Balance)	1	Range Gate Length Control
2	Sweep Amplifier Level Controls (N/S and E/W)	1	Strobe Intensity Control
2	Sweep Amplifier Gain Controls (N/S and E/W)	1	Strobe Slope Control
2	PPI Circularity Controls (N/S and E/W)	1	Strobe Zero Control
1	Cursor Circularity Control (N/S)	1	LAB. Intensity Control
2	D.C. Zero controls (N/S and E/W)	1	Sonobuoy Intensity Control
1	Sweep Speed Control	1	H. M. Intensity Control

3.3.7.6 Cathode Ray Tube - The cathode ray tube shall be a 7 inch PPI type P7 using a motor-stator type deflection yoke. The focus assembly shall be an electromagnetic coil with a focus current control.

3.3.7.7 Deflection Yoke - The deflection yoke retentivity and permeability shall be such that open centering is not visually discernible.

3.3.7.8. High Voltage Supply - The high voltage supply shall be a hermetically sealed, rectified audio frequency supply.

3.3.7.9 Range Circuits - The range sweep generating circuits shall generate the necessary sweeps to be applied to the stator of a two-phase resolver in the synchronizer servo drive asembly, and the stator of a two-phase resolver in Azimuth and Range Indicator IP-203A/APS-20B. The following sweeps shall be provided for the four normal positions of the range switch at 300 pps.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

Switch Position	Sweep Speed	Duration
20 mi/10 mi	20 mi/radius/10 mi/radius	80 mi/80 mi
50 mi	50 mi/radius	200 mi
100 mi	100 mi/radius	200 mi
200 mi	200 mi/radius	200 mi

The same sweep speeds shall be available for the normal positions of the RANGE switch on 900 pps, except that the PPI gate length for any setting of the switch shall be 50 miles. Provision shall be incorporated in the sweep circuits so that as the RANGE switch is changed the physical displacement of the spot will vary so that the displacement in miles remains constant, except for a maximum of a 2-1/2 mile/change in centering of the display when switched to the 10 mile range. A two-position switch shall be provided to select either 10 or 20 mile per radius sweep for the SR (short range) position of the RANGE switch.

3.3.7.10 Normal-Open Center - A NORMAL-OPEN CENTER switch shall be provided on the front panel. On NORMAL position sweep circuits, the range strobe and unblanking circuits will be triggered by the modulator trigger. On OPEN CENTER position, the sweeps will be triggered by the cosine pulse, while the range strobe and unblanking circuits will be triggered by the modulator trigger.

3.3.7.11 Local-Slew - Local slew voltages, set by means of one local slew potentiometer in each rectangular coordinate, shall be capable of moving the sweep starting point ±150 miles in N/S and/or E/W direction. An on-off toggle switch to remove the local slew voltage shall be provided.

3.3.7.12 Slew Mixing Circuits - The mixing circuits shall provide for the algebraic addition of input voltages from the AN/APA-57B and the local slew voltages so that the resultant displacement of the sweep center in miles is the algebraic sum of the aircraft displacement from the operating center and the local slew setting in miles.

3.3.7.13 Strobe - An electronic strobe generated by a range marker, controllable in intensity, with continuous delay from 0 to 200 miles, shall be available for display on all sweeps and the electronic cursor at a range corresponding to the delay setting on a delay counter. Accuracy shall be within 0.5 mile, or 1% of the delay setting, whichever is greater. This marker shall not exceed one microsecond in width at 85% of peak voltage.

3.3.7.14 Range Data Transmission Provisions - A space 1-3/4 inches by 1-3/4 inches by 4-5/8 inches shall be left vacant so that an electromechanical counting switch may be added (if desired) at a later date.

3.3.7.15 Electronic Cursor - A cursor resolver, geared to a bearing crank and counter, shall be provided to receive the output of range sweep circuits and furnish sweeps to the deflection yoke. This cursor sweep shall be time-shared with the main rotating sweep at a ratio of approximately 1 to 12. Provisions shall be made to switch off the time sharing and cursor at minimum cursor intensity. Provisions shall be made so that for centered cursor the start of the cursor sweep may originate within 1/12 tube radius of the main sweep origin, and for slewed cursor the start of the cursor sweep may originate at any coordinate on the display as selected by the N/S and E/W uncalibrated slew controls. Selection of the type cursor operation desired shall be by means of a front panel SLEW-CRT toggle switch. No video except the range strobe shall be displayed on the cursor.

3.3.7.16 Video Amplifier - The video amplifier shall be designed to pass pulses from 0.5 to 1000 microseconds duration with a maximum of 0.25 microsecond rise time and one db "droop". Provisions shall be made to accept and mix singly or in any combination the following video signals: radar or AMTI video; beacon or sonobuoy video; IFF Mode 1; IFF Mode 2; IFF Mode 3; LAB plus HM or HM video. In addition range marks and range strobe shall be included in the PPI display either by mixing with the video signals or by injunction on separate grids of the KRT. Provisions shall be incorporated in the LAB., HM and sonobuoy vider input channels to set the peak video input level at 2 volts. A fast time constant circuit shall be designed to reduce the input coupling circuit time constant to approximately 6 microseconds. Three IFF MODE selector switches shall be provided. In the ON position of these switches the IFF equipment shall be placed in the interrogate condition for the particular mode selection, and the incoming video for that mode shall be connected to the video mixing circuits. A video input channel shall be provided for accepting AMTI video from associated

Source: https://assist.dla.mil -- Downloaded: 202640-17T07:43Z
Check the source to verify that this is the current version before use.

equipment. The connection of this video to the video mixing circuits shall be controlled from the front panel RADAR-OFF-AMTI switch. Range markers shall be included in the PPI display for all positions of the RANGE switch as follows:

20/10 miles - 10 mile range marks
50 miles - 10 mile range marks
100 miles - 20 mile range marks
200 miles - 20 mile range marks

A video input channel shall be provided for accepting sonobuoy video signals from AN/ARR-26 Sonobuoy Receiver equipment, with connection of this video to the mixing circuits controllable from the front panel BEACON-OFF-SONOBUOY video selector switch. A video input channel shall be provided for accepting LAB target tracking marker video from the AN/APA -16 Bombing Computer equipment. A video input channel shall be provided for accepting the heading marker video from Electrical Synchronizer SN-55/APS-20B. A front panel LAB-OFF-HM switch shall select LAB plus HM video or HM video alone.

3.3.7.17. Photographic Coding - Azimuth and Range Indicator IP-203A/APS-20B shall be capable of supplying 3 volts d-c to the coding lamps in Recording Camera CR-1A. A switching network will channel the d-c voltage to the proper coding lamps to correspond to the settings of the AMTI switch and the range switches of Azimuth and Range Indicator IP-203A/APS-20B.

3.3.7.18 Plastic Front Panel - The front panel for the controls shall consist of a plastic lighting plate in accordance with Specification MIL-P-7788. Function and position of all controls shall be visible under all conditions of illumination, except that it will not be required to indicate the position of toggle and push button switches. Toggle switches shall be ON in the up position. A dimmer control shall be provided on the front panel to control the intensity of panel lighting.

3.3.7.19 Reflection Plotter - The cathode ray tube shall be provided with a close fitting, edge lit, engraved lucite or equivalent plastic reflection plotter. The plastic shall have a semi reflection coating #155-49-155 or equivalent on the inside (concave side). The outer edge of the plotting surface shall be engraved with an azimuth scale and the center of the plotting surface shall be marked with an engraved crosshair. The reflection plotter shall be shaped to follow within practical limits the contour of the KRT face and shall reflect 35% ± 10% of the incident light (transmission/reflection ratio equal to 65/35 approximately).

3.3.7.20 Sweep Resolvers - The resolvers used in the sweep circuits of Azimuth and Range Indicator IP-203A/APS-20B and located in both Electrical Synchronizer SN-55/APS-20B and Azimuth and Range Indicator IP-203A/APS-20B shall have 735 ± 3% mh inductance.

3.3.8 Radar Set Control C-1449/APS-20E

3.3.8.1 Function - This unit shall provide means for controlling the transmitter, antenna, receiver and relayed video. It shall provide means for monitoring voltages, currents and significant waveforms for the system with the addition of an auxiliary oscilloscope.

3.3.8.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing EI-237-21. It shall be mounted on Mounting MT-1136/APS-20B.

3.3.8.3 Weight - The weight of Radar Set Control C-1449/APS-20E shall not exceed 30 pounds.

3.3.8.4 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

3.3.8.5 Controls - The location, lettering, and type of all controls shall be in accordance with Bureau of Aeronautics Drawing EI-237-21.

(1) The Main Power control shall be a three-position OFF-STANDBY-OPERATE switch. In the STANDBY position, it shall turn all filaments, apply keep-alive voltages, energize the various blowers, the coolant pump and air pump and run the 3-phase high voltage auto-transformer to its lower limit position, and

Source: https://assist.dla.mil -- Downloaded: 202610-17T07:43Z
Check the source to verify that this is the currentversion before use.

apply all voltages from the low voltage power supply to the system. When the magnetron filament voltage has been raised to a predetermined level, the time delay motor will be energized. In the OPERATE position, it shall apply operating voltages throughout the system, provided the time delay motor has timed out.

(2) The Overload Re-Set control shall be a push button type switch for reapplying operating voltages to the system in the event of persistent modulator overloads or magnetron arc-overs.

(3) The Modulator High Voltage control shall be a three-position RAISE-OUT-LOWER toggle switch for controlling the motor driven three-phase variable auto-transformer in Transformer Assembly TF-158/APS-20B.

(4) The Sector Bearing control shall be a synchro control transformer, type AY-202-3B or equivalent for selecting the center-bearing of a selected sector. The SECTOR width control shall be a potentiometer controlling the width of the antenna sector.

(5) The Manual Tilt Control shall (by means of an AY-202-B autosyn) control the angle of beam tilt to within ±1 degree of the indicated position.

(6) An eight-position-video selector switch which will automatically select the correct trigger for various videos shall be provided to permit selection of the following waveforms for systems test of the radar by the use of an associated standard A and R scope.

Video	Trigger
Radar Video	Modulator Trigger
Beacon Video	Modulator Trigger
Relay Video	Basic Pulse
10-Mi Range Mks	Modulator Trigger
Thyratron Trigger	Basic Pulse
Modulator Trigger	Basic Pulse
Transmitter Pulse	Basic Pulse
Sine-Cosine Pulses	Basic Pulse

(7) Provisions shall be made for controlling the type video provided by the synchronizer for the relay transmitter. It shall be possible to select radar, beacon, and IFF video (as determined by IFF Control C-1040/APX-7), either singly or in any desired combination. In addition to the provisions for controlling the type of video, a relay transmitter power switch shall be provided. The controls shall consist of the following:

(a) Relay Transmitter Power XMTR-OUT Toggle Switch

(b) Beacon Video BCN-OUT Toggle Switch

(c) IFF Video MIX-SHARE Switch

(d) Radar Video-Grid Map RAD-OFF-RAD G. M. Toggle Switch

(8) The RAD-OFF-RAD G. M. toggle switch shall be a three-position toggle switch. In the upper RAD position, radar video will be provided to the synchronizer and thence to the relay transmitter; when in the OFF position no radar will be provided; and when in the RAD G. M. position one section of the switch shall ground a relay circuit from the AN/APA-56 equipment which will provide a grid-map signal to mix with the radar video to the synchronizer.

3.3.8.6 Warning Lights - Three warning lights shall be provided that light under the following system conditions.

Source: https://assist.dla.mil -- Downloaded: 202640-17T07:43Z
Check the source to verify that this is the current version before use.

(a) When air pressure in the pressurized components drops below a safe level.

(b) Failure of magnetron cooling system to operate properly.

(c) Modulator overload and time-out indication.

3.3.8.7 Monitor Meter - A combined a-c/d-c monitoring meter shall be provided. The a-c meter shall be capable of measuring the 3-phase line to neutral voltages, the 400 cycle exact line voltage, and magnetron filament voltage. The d-c meter shall be capable of measuring the d-c line voltage, the modulator high voltage power supply current, and the magnetron anode current.

3.3.8.8 Plastic Front Panel - The front panel of this unit shall be a separate plastic lighting sheet per Specification MIL-P-7788. Function and position of all controls shall be visible under all conditions of illumination, except that it will not be required to indicate the position of toggle and push button switches. Toggle switches shall be ON in the up position. A dimmer control shall be provided on the panel to control the intensity of the panel lighting.

3.3.9 Power Supply PP-347/APS-20B

3.3.9.1 Function - This unit shall supply all operating voltages required by the system except for the modulator and KRT high voltages. It shall furnish the required regulated voltages for the AN/APA-57B corrected fix output potentiometers.

3.3.9.2 Form Factor and Mounting - The overall dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing EI-237-25. It shall be mounted on Mounting MT-1134/APS-20B.

3.3.9.3 Weight - The weight of Power Supply PP-347/APS-20B shall not exceed 65 pounds.

3.3.9.4 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

3.3.9.5 Output Characteristics

Regulation	Voltage in Volts	Voltage Tolerance in Volts	Maximum Allowable Ripple Voltage Peak to Peak in Millivolts	Approximate Current in MA
Regulated	+300	2	0.1	510
Regulated	+152	2	0.1	1000
Regulated	-152	2	0.1	470
Regulated	-255	2.5	0.5	60
Unregulated	400	14	15	40
Unregulated	250	10	10	610
Unregulated	300	10	10	90

3.3.10 Range-Azimuth Indicator IP-23/APS-20B

3.3.10.1 Function - This auxiliary indicator shall provide for PPI presentation on a 5 inch KRT of radar or beacon video on either a 10, 50 or 150 mile sweep.

3.3.10.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing EI-237-27. It shall be mounted directly on the aircraft structure.

3.3.10.3 Weight - The weight of Range-Azimuth Indicator IP-23/APS-20B shall not exceed 15 pounds.

3.3.10.4 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

3.3.10.5 Controls - The location, lettering, and type of all controls shall be in accordance with Bureau of Aeronautics Drawing EI-237-27.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

3.3.10.6 Azimuth Data - Sweep will be positioned relative to the aircraft heading by a fixed motor stator type deflection yoke which accepts resolved sweeps from a one speed, three phase sweep resolver in the antenna. The azimuth bearing indicated on the indicator shall be within 2 degrees of the relative bearing of the antenna. The top of the tube shall indicate aircraft heading. A permanent color filter shall be attached to the front of the cathode ray tube and engraved for azimuth measurements.

3.3.10.7 Visor - A rubber visor shall be provided.

3.3.10.8 Range Marks - Provisions shall be incorporated for accepting 10 and 20 mile range marks from the synchronizer.

3.3.11 Power Supply PP-704/APS-20B

3.3.11.1 Function - This unit shall contain a compact audio oscillator type high voltage power supply to supply the necessary anode and grid voltages to, and the range sweep generating circuits for, Range-Azimuth Indicator IP-23/APS-20B.

3.3.11.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing EI-237-28. It shall be mounted on Mounting MT-954/APS-20B.

3.3.11.3 Weight - The weight of Power Supply PP-704/APS-20B shall not exceed 25 pounds.

3.3.11.4 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

3.3.11.5 Output Voltage & Sweep Characteristics - Ripple shall be limited in the output in such a manner as to have no discernible effect on the KRT presentation. Both 50 and 150 mile sweep generating circuits shall be provided with selection at Range-Azimuth Indicator IP-23/APS-20B. Suitable circuits shall be provided for blanking the KRT, clamping the deflection amplifiers and gating the sweep circuits. Provision shall be incorporated so that the unblanking gate will be automatically switched to 50 miles when the radar is operating on high repetition rate.

3.3.12 Transformer Assembly TF-158/APS-20B

3.3.12.1 Function - This unit contains the variable auto-transformer that provides three phase voltage to Power Supply PP-829/APS-20B, one phase voltage for the magnetron filament, the control chassis which regulates the magnetron filament voltage, and the main power contactor.

3.3.12.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing EI-237-18. It shall be mounted directly on the aircraft structure.

3.3.12.3 Weight - The weight of Transformer Assembly TF-158/APS-20B shall not exceed 80 pounds.

3.3.12.4 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

3.3.12.5 Contents of Unit

(1) The three phase auto-transformer shall be driven by a 26.5 volt d-c gear motor. Adjustable limit switches shall be provided to de-energize the drive motor by preset maintenance adjustments at upper and lower voltage limits.

(2) The single phase auto-transformer shall be driven by a 26.5 volt d-c motor and shall provide the magnetron filament voltage.

(3) The control chassis shall regulate the filament voltage within ±2.5% over the frequency range from 380 to 1000 cps.

Source: https://assist.dla.mil -- Downloaded: 202640-17T07:43Z
Check the source to verify that this is the current version before use.

3.3.13 Directional Coupler CU-316/APS-20B

3.3.13.1 Function - The Directional Coupler shall provide for coupling a fixed percentage of the transmitter power into an auxiliary line. The directivity shall be 20 db minimum. The coupling shall be 40 ± 2 db. The VSWR shall not exceed 1:12. The VSWR of the type "N" auxiliary fitting shall not exceed 1:3.

3.3.13.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing EI-237-6. This unit shall be mounted as part of the r-f transmission line.

3.3.13.3 Weight - The weight of Directional Coupler CU-316/APS-20B shall not exceed 4 pounds.

3.3.13.4 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

3.3.14 Gyroscope CN-219A/APS-20B

3.3.14.1 Function - The gyro assembly shall provide a vertical reference for the positioning of the radar beam during stabilized operation. The gyro vertical shall agree with the true vertical within an accuracy of ±1 degree. This accuracy shall be maintained for tilt of 15 degrees or less, tilt rates of 32 deg/sec or less,and tilt accelerations of 50 deg/sec/sec or less.

3.3.14.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing EI-237-4. It shall be mounted directly on the aircraft structure.

3.3.14.3 Contents of Unit - This unit shall contain a gyroscope and its erection amplifier mounted on a common base. This unit shall be marked with a forward position for the purpose of establishing an electrical alignment reference.

3.3.14.4 Weight - The weight of Gyroscope CN-219A/APS-20B shall not exceed 25 pounds.

3.3.14.5 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

3.3.15 Gyroscope CN-218/APS-20B

3.3.15.1 Function - This gyroscope shall operate a switch to cut out roll erection of the vertical gyro when the aircraft is in a turn.

3.3.15.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing EI-237-5. It shall be mounted directly on the aircraft structure.

3.3.15.3 Weight - The weight of Gyroscope CN-218/APS-20B shall not exceed 2 pounds.

3.3.15.4 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

3.3.16 Air Compressor HD-126/APS-20B

3.3.16.1 Function - The Air Compressor shall provide and maintain pressurization of the r-f plumbing, Power Supply PP-829/APS-20B and Radar Transmitter T-467/APS-20E.

3.3.16.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing EI-237-31. It shall be mounted directly on the aircraft structure.

3.3.16.3 Contents of Unit - The pump shall be capable of maintaining the pressurized r-f plumbing between 30 and 35 psi absolute under all operating conditions specified. The transmitter and power supply shall be maintained between 16 and 24 psi absolute by means of a differential value as shown on Bureau of Aeronautics Drawing EI-237-32.

3.3.16.4 Weight - The weight of Air Compressor HD-126/APS-20B shall not exceed 12 pounds.

Source: https://assist.dla.mil -- Downloaded: 20260-17T07:43Z
Check the source to verify that this is the current version before use.

3.3.16.5 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

3.3.17 Dehydrator HD-127/APS-20B

3.3.17.1 Function - The Dehydrator shall contain an activated dessicant for maintaining dry air in the pressurized components of the system.

3.3.17.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing EI-237-30. It shall be mounted directly on the aircraft structure.

3.3.17.3 Contents of Unit - The dehydrator shall be so constructed that, when filled with dessicant, and when the inlet and outlet are sealed, the dessicant will remain activated for a period of time not less than 90 days.

3.3.17.4 Weight - The weight of Dehydrator HD-127/APS-20B shall not exceed 4 pounds.

3.3.18 Cooler, Liquid Electron Tube HD-125/APS-20B

3.3.18.1 Function - The cooling assembly shall be a unit capable of supplying sufficient cooling fluid to the magnetron to maintain the anode temperature below the specified maximum, and shall be capable of dissipating the heat transferred from the magnetron to the cooling fluid.

3.3.18.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing EI-237-33. It shall be mounted on Mounting MT-1137/APS-20B.

3.3.18.3 Contents of Unit - The cooling fluid used in the radiating assembly shall be nontoxic, non-inflammable, non-corrosive, chemically stable, and having a freezing and boiling point consistent with the operating requirements of the QK-428 magnetron and the service operating conditions specified therein.

3.3.18.4 Weight - The weight of Cooler HD-125/APS-20B shall not exceed 32.5 pounds.

3.3.18.5 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

3.3.19 Amplifier, Electronic Control, AM-839/APS-20B

3.3.19.1 Function - The electronic control amplifier shall receive the antenna tilt error signal and amplify it. The output of the electronic control amplifier shall control the magnetic amplifier which in turn shall energize the servo motor to position the radar beam. In addition to displacement control, such derivative and integral control as may be required to meet this specification shall be incorporated.

3.3.19.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing EI-237-9. It shall be mounted on Mounting MT-802/APS-20B.

3.3.19.3 Weight - The weight of Amplifier, Electronic Control, AM-839/APS-20B shall not exceed 4.5 pounds.

3.3.19.4 Electrical Connections - External electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

3.3.20 Amplifier, Magnetic AM-767/APS-20B

3.3.20.1 Function - The output of Amplifier, Electronic Control, AM-839/APS-20B shall control Amplifier, Magnetic, AM-767/APS-20B. The output of Amplifier Magnetic AM-767/APS-20B shall control the two phase servo motor to position the radar beam in elevation.

3.3.20.2 Form Factor and Mounting - The outline dimensions shall not exceed the dimensions shown on Bureau of Aeronautics Drawing EI-237-10. It shall be mounted on Mounting MT-802/APS-20B.

3.3.20.3 Weight - The weight of Amplifier, Magnetic, AM-767/APS-20B shall not exceed 8 pounds.

3.3.20.4 Electrical Connections - The external electrical connections shall be in accordance with Bureau of Aeronautics Drawing EI-237-36.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests - The testing and inspection of aircraft electronic equipment shall be classified as follows:

(1) Preproduction Tests - Preproduction tests are those conducted on samples which are representative of the production equipment to be supplied under the contract, for the purpose of determining that the equipment meets all the requirements of the applicable specifications. Preproduction tests shall consist of the following two tests:

(a) Contractor Demonstration Tests - Contractor Demonstration Tests are accomplished under the responsibility of the contractor on a sample representative of the production equipment, to demonstrate that the equipment complies with all the requirements of the applicable specifications.

(b) Service Approval Tests - Service Approval Tests are accomplished under the responsibility of the procuring service on a sample representative of the production equipment, to determine that the equipment meets all the requirements of the specification. These tests may be accomplished by having the procuring service supervise the Contractor Demonstration Tests.

(2) Inspection Tests - Inspection Tests are those tests accomplished on equipment submitted for acceptance under the contract. Inspection Tests shall consist of the following:

(a) Individual Tests - Individual Tests are those tests conducted on each equipment submitted for acceptance under the contract.

(b) Sampling Tests - Sampling Tests are those tests accomplished on a quantity of equipment as approved by the Bureau of Aeronautics and which have previously passed the Individual Tests.

4.2 Preproduction Tests

4.2.1 Contractor Demonstration Tests - The contractor shall conduct tests on one or more sample equipments as necessary to determine that the design of the equipment as proposed by the contractor will meet the requirements of all applicable specifications. Contractor Demonstration Tests shall be conducted in accordance with the approved preproduction test procedure. The data obtained by the contractor in conducting these tests shall be included with the design data submitted with the Service Approval Test sample. The Inspector of Naval Material and the Bureau of Aeronautics shall be advised when tests are to be conducted so that the Bureau of Aeronautics may designate a representative to witness or supervise the test when so desired. When tests are supervised the Bureau of Aeronautics may decide to omit certain portions of the Service Approval Tests which will duplicate the Contractor Demonstration Tests, however, in conducting Service Approval Tests the Bureau of Aeronautics may elect to repeat any test previously conducted by the contractor, when deemed necessary. Contractors not having laboratory facilities to satisfactorily conduct all tests shall obtain either the services of a commercial testing laboratory or written approval from the Bureau of Aeronautics to omit that portion of the tests which cannot be conducted.

4.2.2 Service Approval Tests - The Service Approval sample shall consist of one equipment representative of the production equipment to be supplied under the contract. This equipment shall be delivered by the contractor to the destination specified by the Bureau of Aeronautics. The sampling procedure outlined in Specification MIL-P-7788 shall apply for plastic lighting plates.

4.2.2.1 Accessory Materials Required - In addition to the complete equipment submitted, the contractor shall submit the accessory material and design data as specified in Specification MIL-E-5400.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.

4.2.3 Scope of Tests - Preproduction Tests shall include all tests deemed necessary to determine that the model meets all the requirements of this specification and the contract. Preproduction Tests shall include testing in accordance with Specification MIL-T-5422 and Specification MIL-I-6181. The testing of plastic lighting plates shall be in accordance with Specification MIL-P-7788. The design and test data supplied by the contractor shall indicate the equipment's physical and electrical characteristics and establish that the equipment meets all the requirements of this specification. The test data shall include data obtained by the contractor in testing to MIL-T-5422.

4.3 Inspection Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the required tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government Inspector shall engage the services of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all inspection tests required by this specification, and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Inspection tests shall consist of the following:

(1) Individual Tests - Individual tests shall be conducted on each equipment submitted for acceptance under the contract.

(2) Sampling Tests - Sampling tests shall be conducted on a quantity of equipment as specified in 4.3.2.1. Equipments for sampling tests shall be selected by the Government Inspector after the equipment has passed the individual tests and has been packaged and packed ready for shipment.

4.3.1 Individual Tests - Each equipment shall be examined carefully to determine compliance with the requirements of this specification with respect to material and workmanship. In addition, the equipment shall be given an operational check, with sufficient data recorded to assure satisfactory operation.

4.3.1.1 Scope of Test - As a minimum each equipment shall be tested to determine compliance with the following under standard laboratory conditions and nominal input voltages.

	Individual Tests	Applicable Paragraph
I	AN/APS-20E System	
	(1) Warm up time	3.2.4
	(2) Stability (4 hour check only)	3.2.4
	(3) Input Power	3.2.8
	(4) System Noise Figure	3.2.18.2
	(5) System Recovery Time	3.2.18.6
II	Antenna AS-407/APS-20B	
	(1) Antenna Pressure Leakage & Safety Test	3.3.1.5
	(2) Radar Voltage Standing Wave Ratio	3.3.1.7.1
	(3) IFF Voltage Standing Wave Ratio	3.3.1.7.1
	(4) Azimuth Rotation Speeds	3.3.1.10.1
	(5) Azimuth Sector Scan	3.3.1.10.2
	(6) Tilt Stabilization Accuracy	3.3.1.14.1
	(7) Tilt Manual Control Accuracy	3.3.1.14.2
	(8) Tilt Limit Operation	3.3.1.14.3
	(9) Tilt Stow Operation	3.3.1.14.4
	(10) Tilt Information Accuracy	3.3.1.14.5
III	Duplexer CU-165/APS-20B	
	(1) Duplexer pressure leakage and safety test	3.3.2.6
	(2) Frequency of beacon filter cavity	3.3.2.12
	(3) Crystal shutter attenuation	3.3.2.13

Source: https://assist.dla.mil -- Downloaded: 202640-17T07:43Z
Check the source to verify that this is the currentversion before use.

IV	Receiver R-251/APS-20B	
	(1) Radar IF Frequency	3.3.3.6.1.1
	(2) IAGC Response	3.3.3.6.1.2
	(3) Radar IF bandwidth	3.3.3.6.1.3
	(4) AMTI bandwidth	3.3.3.6.1.3
	(5) Radar STC attenuation	3.3.3.6.1.5
	(6) Radar LO frequency	3.3.3.6.2.1
	(7) AFC operation	3.3.3.6.2.2
	(8) Radar Video rise time & droop	3.3.3.6.3.1 & 2
	(9) Beacon IF frequency	3.3.3.7.1.1
	(10) Beacon IF bandwidth	3.3.3.7.1.2
	(11) Beacon LO frequency	3.3.3.7.2.1
	(12) Tuning range of beacon reference cavity	3.3.3.7.2.1
	(13) Beacon video rise time & droop	3.3.3.7.3.1
V	Power Supply PP-829/APS-20E	
	(1) Pressure leakage and safety	3.3.4.5
	(2) Output voltage, current & ripple	3.3.4.6
VI	Radar Transmitter T-467/APS-20E	
	(1) Transmitter pressure leakage & safety test	3.3.5.5
	(2) Interlock protection	3.3.5.6
	(3) Power output	3.3.5.8.1
	(4) Current pulse width	3.3.5.8.2
	(5) Transmitter frequency	3.3.5.8.4
VII	Electrical Synchronizer SN-55/APS-20B	
	(1) Basic pulse amplitude	3.3.6.5
	(2) Repetition frequencies	3.3.6.5
	(3) Sine pulse spacing	3.3.6.6
	(4) Cosine pulse spacing	3.3.6.7
	(5) Modulator trigger amplitude & position	3.3.6.8
	(6) IFF trigger amplitude	3.3.6.9
	(7) Coded relay information	3.3.6.10
	(8) Coding of IFF pulses	3.3.6.11
	(9) Video amplitude and mixing	3.3.6.12
	(10) Range Mark amplitude, duration & spacing	3.3.6.13
	(11) Heading Mark accuracy	3.3.6.14
	(12) Azimuth servo accuracy	3.3.6.15
	(13) Azimuth sector width	3.3.6.16
VIII	Azimuth and Range Indicator IP-203A/APS-20B	
	(1) All sweep speeds & duration	3.3.7.9
	(2) Open center operation	3.3.7.10
	(3) Local slew operation	3.3.7.11
	(4) APA-57 slew operation	3.3.7.12
	(5) Strobe accuracy	3.3.7.13
	(6) Cursor display	3.3.7.15
	(7) Video rise time and droop	3.3.7.16
IX	Power Supply PP-347/APS-20B	
	(1) Voltage and ripple output	3.3.9.5
X	Range-Azimuth Indicator IP-23/APS-20B	
	(1) Presentation	3.3.10.1
	(2) Azimuth accuracy	3.3.10.7
XI	Power Supply PP-704/APS-20B	
	(1) Sweep speed and duration	3.3.11.5

Source: https://assist.dla.mil -- Downloaded: 202610-17T07:43Z
Check the source to verify that this is the current version before use.

4.3.2 Sampling Tests - Sampling Tests shall be conducted under the close supervision of the Government Inspector and shall include a performance check at ambient laboratory conditions. This Test may also include any of the other preproduction tests the Inspector deems necessary to determine conformance of the equipment with the requirements of this Specification. These Tests shall include also a careful examination of the way the equipment is packaged and packed to determine conformance with the packaging and packing requirements of this specification or the contract. Sampling Tests shall be conducted on complete equipments. When individual units of the equipments are tested individually and shipped then the samples selected for Sampling Tests shall consist of all the individual units necessary to comprise a complete operating equipment.

4.3.2.1 Sampling Test Schedule - Sampling Tests shall be performed on the following quantity of equipment unless otherwise stated in the contract or approved in writing by the procuring service.

Quantity of Equipment	Quantity of Equipment to be tested
First 10 equipments	1
Next 10 to 100	2
Next 100 to 200	1
Over 200	One for each additional 200

4.3.2.2 Scope of Test - As a minimum, each equipment selected for sampling test shall be tested to determine compliance with the following requirements of this specification using MIL-T-5422 and MIL-I-6181 as Procedure Specifications where applicable.

	Sampling Tests	Applicable Paragraph
I	AN/APS-20E System	
	(1) Warm up time	3.2.4
	(2) Stability (4 hour check only)	3.2.4
	(3) Input Power	3.2.8
	(4) System Noise Figure	3.2.18.2
	(5) System Recovery Time	3.2.18.6
	(6) Conducted & Radiated Noise & Susceptance	3.2.16
II	Antenna AS-407/APS-20B	
	(1) Antenna Pressure Leakage & Safety Test	3.3.1.5
	(2) Radar Voltage Standing Wave Ratio	3.3.1.7.1
	(3) IFF Voltage Standing Wave Ratio	3.3.1.7.2
	(4) Antenna Radiation Patterns	3.3.1.9
	(5) Azimuth Rotation Speeds	3.3.1.10.1
	(6) Azimuth Sector Scan	3.3.1.10.2
	(7) Tilt Stabilization Accuracy	3.3.1.14.1
	(8) Tilt Manual Control Accuracy	3.3.1.14.2
	(9) Tilt Limit Operation	3.3.1.14.3
	(10) Tilt Stow Operation	3.3.1.14.4
	(11) Tilt Information Accuracy	3.3.1.14.5
III	Duplexer CU-165/APS-20B	
	(1) Duplexer pressure leakage and safety test	3.3.2.6
	(2) Frequency of beacon filter cavity	3.3.2.12
	(3) Crystal shutter attenuation	3.3.2.13
IV	Receiver R-251/APS-20B	
	(1) Radar IF Frequency	3.3.3.6.1.1
	(2) IAGC Response	3.3.3.6.1.2
	(3) Radar IF bandwidth	3.3.3.6.1.3
	(4) AMTI bandwidth	3.3.3.6.1.3
	(5) Radar STC attenuation	3.3.3.6.1.5
	(6) Radar LO frequency	3.3.3.6.2.1
	(7) AFC operation	3.3.3.6.2.2

Source: https://assist.dla.mil -- Downloaded: 202640-17T07:43Z
Check the source to verify that this is the currentversion before use.

IV	Receiver R-251/APS-20B (Cont'd)	
	(8) Radar Video rise time & droop	3.3.3.6.3.1 & 2
	(9) Beacon IF frequency	3.3.3.7.1.1
	(10) Beacon IF bandwidth	3.3.3.7.1.2
	(11) Beacon LO frequency	3.3.3.7.2.1
	(12) Tuning range of beacon reference cavity	3.3.3.7.2.1
	(13) Beacon video rise time & droop	3.3.3.7.3.1
V	Power Supply PP-829/APS-20B	
	(1) Pressure leakage and safety	3.3.4.5
	(2) Output voltage, current & ripple	3.3.4.6
VI	Radar Transmitter T-467/APS-20E	
	(1) Transmitter pressure leakage & safety test	3.3.5.5
	(2) Interlock protection	3.3.5.6
	(3) Power output	3.3.5.8.1
	(4) Current pulse width	3.3.5.8.2
	(5) Transmitter frequency	3.3.5.8.4
VII	Electrical Synchronizer SN-55/APS-20B	
	(1) Basic pulse amplitude	3.3.6.5
	(2) Repetition frequencies	3.3.6.5
	(3) Sine pulse spacing	3.3.6.6
	(4) Cosine pulse spacing	3.3.6.7
	(5) Modulator trigger amplitude & position	3.3.6.8
	(6) IFF trigger amplitude	3.3.6.9
	(7) Coded relay information	3.3.6.10
	(8) Coding of IFF pulses	3.3.6.11
	(9) Video amplitude and mixing	3.3.6.12
	(10) Range Mark amplitude, duration & spacing	3.3.6.13
	(11) Heading Mark accuracy	3.3.6.14
	(12) Azimuth servo accuracy	3.3.6.15
	((13) Azimuth sector width	3.3.6.16
VIII	Azimuth and Range Indicator IP-203A/APS-20B	
	(1) All sweep speeds & duration	3.3.7.9
	(2) Open center operation	3.3.7.10
	(3) Local slew operation	3.3.7.11
	(4) APA-57 slew operation	3.3.7.12
	(5) Strobe accuracy	3.3.7.13
	(6) Cursor display	3.3.7.15
	(7) Video rise time and droop	3.3.7.16
IX	Power Supply PP-347/APS-20B	
	(1) Voltage and ripple output	3.3.9.5
X	Range-Azimuth Indicator IP-23/APS-20B	
	(1) Presentation	3.3.10.1
	(2) Azimuth accuracy	3.3.10.7
XI	Power Supply PP-704/APS-20B	
	(1) Sweep speed and duration	3.3.11.5

4.3.2.3 Equipment Failure - Should an equipment fail to pass any part of the Sampling Test, the following corrective action shall be taken by the contractor.

(1) Determine the cause of the failure.

(2) Determine if the failure is an isolated case or a design defect.

(3) If a design defect, suggest corrective action for BuAer approval.

(4) Include a test in the Individual Test to check all equipments for this requirement until reasonable assurance can be obtained that the defect has been satisfactorily corrected.

Source: https://assist.dla.mil -- Downloaded: 202610-17T07:43Z
Check the source to verify that this is the current version before use.

4.4 Test Procedures - The procedures and methods for conducting Pre-production Tests and Inspection Tests shall be prepared by the contractor and sent to the Bureau of Aeronautics for approval. The right is reserved by the Bureau of Aeronautics or the Naval Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Use MIL-T-18303 as a guide for preparation of Test procedures.

4.5 Pre-Submission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply to the best of his knowledge and belief with all applicable requirements. The design data submitted by the contractor for the pre-production model shall indicate its physical and electrical characteristics and other necessary test data which will establish that the equipment meets all the requirements of this specification.

4.6 Preproduction Sample Approval - Approval of the preproduction sample shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No additional equipments shall be delivered prior to the approval of the pre-production sample. Prefabrication of any equipments prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of the equipment to be submitted for acceptance under the contract. The preproduction sample shall not be considered as one of the equipments under the contract, however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be packaged, packed and marked for shipment in accordance with the requirements of Specifications JAN-P-658 and MIL-STD-129.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 Vd-c and 115 ± 1 Va-c. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

PATENT NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 202610-17T07:43Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4063, WHICH IS OBSOLETE PLATE NO. 16419

Source: https://assist.dla.mil -- Downloaded: 202610-17T07:43Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

FOLD

PLATE NO. 15419 (BACK

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T07:43Z
Check the source to verify that this is the current version before use.